Homework 8

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Section 3.2

3.2–8

Given $X \sim \text{Gamma}(\alpha = 2,\ \theta = 4)$. Compute $P(X < 5)$.

For $\alpha = 2$ (shape-scale form) the CDF is

$$F(x) = 1 - e^{-x/\theta}\left(1 + \frac{x}{\theta}\right).$$

Plug in $x = 5,\ \theta = 4$:

$$\frac{x}{\theta} = \frac{5}{4} = 1.25, \qquad F(5) = 1 - e^{-1.25}(1 + 1.25) = 1 - e^{-1.25}(2.25).$$

Use $e^{-1.25} \approx 0.2865$:

$$F(5) \approx 1 - (0.2865)(2.25) = 1 - 0.6446 \approx 0.3554.$$

$$\boxed{P(X < 5) \approx 0.355}$$

3.2–12

Poisson process with rate 16 counts/sec. Let W be time until the 7th count.

(a) Distribution: Waiting time for the kth event in a Poisson process is Gamma with shape k and scale $1/\lambda$. So

$$W \sim \text{Gamma}(\alpha = 7,\ \theta = 1/16).$$

(b) Compute $P(W \le 0.5)$.

Using the relation for the kth arrival:

$$P(W \le w) = 1 - e^{-\lambda w} \sum_{n=0}^{k-1} \frac{(\lambda w)^n}{n!}.$$

Here $\lambda w = 16(0.5) = 8$, $k = 7$, so

$$P(W \le 0.5) = 1 - e^{-8} \sum_{n=0}^{6} \frac{8^n}{n!}.$$

Compute the finite sum (terms shown):

$$\sum_{n=0}^{6} \frac{8^n}{n!} = 1 + 8 + 32 + 85.3333 + 170.6667 + 273.0667 + 364.0889 \approx 934.1566.$$

With $e^{-8} \approx 0.0003354626$:

$$P(W \le 0.5) \approx 1 - (0.0003354626)(934.1566) \approx 1 - 0.3133 \approx 0.6867.$$

$$\boxed{P(W \le 0.5) \approx 0.687}$$

3.2–13

Let $X \sim \chi^2(23)$.

(a) $P(14.85 < X < 32.01) = F_X(32.01) - F_X(14.85)$. Using $\mathrm{X}^2 CDF values, F_X(14.85) \approx 0.10, \quad F_X(32.01) \approx 0.90$, so

$$\boxed{P(14.85 < X < 32.01) \approx 0.80}$$

(b) Find a, b with $P(a < X < b) = 0.95$ and $P(X < a) = 0.025$.
These are the 2.5% and 97.5% quantiles:

$$a = \chi^2_{0.025}(23) \approx 11.69, \qquad b = \chi^2_{0.975}(23) \approx 38.076.$$

(c) Mean and variance:

$$\mathbb{E}[X] = k = 23, \qquad \mathrm{Var}(X) = 2k = 46.$$

(d) The 5% and 95% chi-square critical values:

$$\chi^2_{0.05}(23) \approx 13.09, \qquad \chi^2_{0.95}(23) \approx 35.17.$$

3.2–16

Cars arrive at rate 5 cars per 10 min $\Rightarrow \lambda = 0.5$ cars/min. Time until the 8th arrival T is Gamma with shape $k = 8$ and scale $\theta = 1/\lambda = 2$.

We want $P(T > 26.30)$. Using the Poisson-process complementary form:

$$P(T > t) = e^{-\lambda t} \sum_{n=0}^{k-1} \frac{(\lambda t)^n}{n!}.$$

Here $\lambda t = 0.5(26.30) = 13.15$ and $k = 8$ so sum $n = 0$ to 7. Evaluating (using a calculator for the sum and exponential) gives

$$P(T > 26.30) \approx 0.05.$$

$$\boxed{P(T > 26.30) \approx 0.05}$$

Section 3.3

3.3–1

Let $Z \sim N(0, 1)$. Use $\Phi(\cdot)$ for the standard normal CDF and explicit Z-table values.

(a) $P(0.47 < Z \leq 2.13) = \Phi(2.13) - \Phi(0.47)$. From Z-table: $\Phi(2.13) = 0.9834$, $\Phi(0.47) = 0.6808$.

$$P = 0.9834 - 0.6808 = 0.3026.$$

(b) $P(-0.97 \le Z < 1.27) = \Phi(1.27) - \Phi(-0.97)$. Z-table: $\Phi(1.27) = 0.8980$, $\Phi(-0.97) = 0.1660$.

$$P = 0.8980 - 0.1660 = 0.7320.$$

(c) $P(Z > -1.56) = 1 - \Phi(-1.56)$. Z-table: $\Phi(-1.56) = 0.0594$.

$$P = 1 - 0.0594 = 0.9406.$$

(d) $P(Z > 2.78) = 1 - \Phi(2.78)$. Z-table: $\Phi(2.78) = 0.9973$.

$$P = 1 - 0.9973 = 0.0027.$$

(e) $P(|Z| < 1.96) = \Phi(1.96) - \Phi(-1.96)$. Z-table: $\Phi(1.96) = 0.9750$, $\Phi(-1.96) = 0.0250$.

$$P = 0.9750 - 0.0250 = 0.9500.$$

(f) $P(|Z| < 1) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = 0.6826$.

(g) $P(|Z| < 2) = \Phi(2) - \Phi(-2) = 0.9772 - 0.0228 = 0.9544$.

(h) $P(|Z| < 3) = \Phi(3) - \Phi(-3) = 0.9987 - 0.0013 = 0.9974$.

3.3–3

Find c in each part (use symmetry/quantiles of standard normal).

(a) $P(Z \ge c) = 0.025 \Rightarrow \Phi(c) = 0.975 \Rightarrow c = 1.96$.

(b) $P(|Z| \le c) = 0.95 \Rightarrow$ each tail $0.025 \Rightarrow \Phi(c) = 0.975 \Rightarrow c = 1.96$.

(c) $P(Z > c) = 0.05 \Rightarrow \Phi(c) = 0.95 \Rightarrow c = 1.645$.

(d) $P(|Z| \le c) = 0.90 \Rightarrow$ each tail $0.05 \Rightarrow \Phi(c) = 0.95 \Rightarrow c = 1.645$.

3.3–5

Let $X \sim N(\mu = 6,\ \sigma^2 = 25)$ so $\sigma = 5$. Use $Z = \dfrac{X - 6}{5}$.

(a) $P(6 \le X \le 14) = P\big(0 \le Z \le \frac{14-6}{5}\big) = \Phi(1.6) - \Phi(0)$. Z-table: $\Phi(1.6) = 0.9452$, $\Phi(0) = 0.5000$.

$$P = 0.9452 - 0.5000 = 0.4452.$$

(b) $P(4 \le X \le 14) = P\big(-0.4 \le Z \le 1.6\big) = \Phi(1.6) - \Phi(-0.4)$. Z-table: $\Phi(-0.4) = 0.3446$.

$$P = 0.9452 - 0.3446 = 0.6006.$$

(c) $P(-4 < X \le 0) = P\big(-2 < Z \le -1.2\big) = \Phi(-1.2) - \Phi(-2)$. Z-table: $\Phi(-1.2) = 0.1151$, $\Phi(-2) = 0.0228$.

$$P = 0.1151 - 0.0228 = 0.0923.$$

(d) $P(X > 15) = P\big(Z > 1.8\big) = 1 - \Phi(1.8)$. Z-table: $\Phi(1.8) = 0.9641$.

$$P = 1 - 0.9641 = 0.0359.$$

(e) $P(|X - 6| < 5) = P(1 < X < 11) = P(-1 < Z < 1) = \Phi(1) - \Phi(-1)$. Z-table: $\Phi(1) = 0.8413,\ \Phi(-1) = 0.1587$.

$$P = 0.8413 - 0.1587 = 0.6826.$$

(f) $P(|X - 6| < 10) = P(-4 < X < 16) = P(-2 < Z < 2) = \Phi(2) - \Phi(-2)$. Z-table: $\Phi(2) = 0.9772,\ \Phi(-2) = 0.0228$.

$$P = 0.9772 - 0.0228 = 0.9544.$$

(g) $P(|X - 6| < 15) = P(-9 < X < 21) = P(-3 < Z < 3) = \Phi(3) - \Phi(-3)$. Z-table: $\Phi(3) = 0.9987,\ \Phi(-3) = 0.0013$.

$$P = 0.9987 - 0.0013 = 0.9974.$$

(h) $P(|X - 6| < 12.4) = P(-6.4 < X < 18.4) = P(-2.48 < Z < 2.48)$. Z-table: $\Phi(2.48) = 0.9934,\ \Phi(-2.48) = 0.0066$.

$$P = 0.9934 - 0.0066 = 0.9868.$$

3.3–11

Mint weights: $X \sim N(21.37,\ 0.16)$ so $\sigma = \sqrt{0.16} = 0.4$.

(a) $P(X > 22.07)$.

Compute

$$Z = \frac{22.07 - 21.37}{0.4} = \frac{0.70}{0.4} = 1.75.$$

Z-table: $\Phi(1.75) = 0.9599$.

$$P(X > 22.07) = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401.$$

(b) 15 mints selected. Let Y be the number with weight < 20.857 g.

First get single-mint probability:

$$Z = \frac{20.857 - 21.37}{0.4} = \frac{-0.513}{0.4} \approx -1.2825.$$

Using the Z-table (rounded to two decimals), $\Phi(-1.28) \approx 0.1003$. So

$$p \approx 0.1003.$$

Now $Y \sim \text{Binomial}(n = 15, p = 0.1003)$. We want $P(Y \le 2) = \sum_{k=0}^{2} \binom{15}{k} p^k (1-p)^{15-k}$.

Compute (rounded):

$$P(0) = (1-p)^{15} \approx 0.2059,$$

$$P(1) = 15p(1-p)^{14} \approx 0.3432,$$

$$P(2) = \binom{15}{2} p^2 (1-p)^{13} \approx 0.2679.$$

Sum:

$$P(Y \le 2) \approx 0.2059 + 0.3432 + 0.2679 = 0.8170.$$

$$\boxed{P(Y \le 2) \approx 0.817}$$